

PHIRAPHAT BOONSODAKORN

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EDUCATION

King Mongkut's Institute of Technology Ladkrabang (KMITL) Bangkok, Thailand

Bachelor of Engineering in IoT System and Information Engineering

Expected Graduation: May 2028

GPA: 2.99 / 4.00

PROJECTS

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| IoT Portable Device for Saline Water Monitoring | Jan 2026 – Present |
| <ul style="list-style-type: none">Developed, as part of a 5-person team, a portable IoT monitoring system for measuring water salinity, recording GPS locations, and transmitting field measurements to a cloud-based platform.Designed the device workflow and control logic covering sensor acquisition, automatic/manual calibration, temperature compensation, salinity calculation, and data transmission.Contributed to system validation; the completed system achieved an average measurement error below 5% and supported remote monitoring via a web dashboard and Telegram. | |
| IoT Department Official Website | Dec 2025 – Mar 2026 |
| <ul style="list-style-type: none">Designed responsive UI/UX prototypes in Figma and implemented the frontend using Vue.js, HTML, and CSS.Converted UI prototypes into reusable Vue.js components for presenting departmental information. | |
| Smart Residential Parcel Management Locker | Aug 2025 – Oct 2025 |
| <ul style="list-style-type: none">Developed, as part of a 2-person team, an IoT parcel management system connecting a web application with Raspberry Pi-controlled lockers for automated parcel drop-off and pickup.Contributed to the implementation of a non-blocking Flask REST API using multithreading and busy-state management to prevent duplicate hardware commands during concurrent requests.Integrated Google Apps Script and Google Sheets with the hardware workflow to support web-triggered locker access, real-time activity logging, and automatic local CSV backup. | |
| AI-Based Parking Space Detection System | Jan 2024 – Mar 2024 |
| <ul style="list-style-type: none">Developed a Python-based computer vision system using YOLO and OpenCV to detect vehicles from CCTV footage and classify predefined parking spaces as available or occupied.Integrated real-time availability notifications and tested the detection workflow using publicly accessible CCTV footage from different real-world environments. | |
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TECHNICAL SKILLS

- Programming:** Python, C/C++, JavaScript, HTML/CSS, SQL
- Frameworks & Libraries:** Vue.js, Flask, OpenCV, YOLO
- IoT & Embedded:** Raspberry Pi, ESP32, Arduino, Sensors
- Tools & Platforms:** Git, GitHub, Figma, Google Sheets API
- APIs & Protocols:** RESTful APIs, HTTP/HTTPS, JSON